DIGITAL EGYPT PIONEERS INITIATIVE (DEPI) — ROUND 5

LUMORA LEARNING PLATFORM

Phase 4: System Analysis & Design — Section 5: System Deployment & Integration

Project Attribute	Specification Details
Project Title	Lumora Learning Platform (Enterprise E-Learning Ecosystem)
Team Name	Code Raiders (6 Members)
Program & Track	DEPI Round 5 — Full-Stack .NET Web Development
Training Partner & Group	AST — Group Code: CAI5_SWD5_S6
Target Cloud Environment	Microsoft Azure App Services, Azure SQL Database, Azure Blob Storage & Azure CDN
Document Deliverable	Phase 4: System Analysis & Design — Part 5: System Deployment & Integration

Team Roster & Integration Ownership Matrix (Code Raiders)

#	Member Full Name	DEPI ID	Deployment & System Integration Responsibility
1	Mohamed Mostafa Sileem Ibrahim (Team Lead)	21161803	Cloud Architect & DevOps Lead: Azure App Service Provisioning, GitHub Actions CI/CD Pipeline, DI Container & N-Tier Assembly Integration
2	Osama Mamdouh Saber Abdelnaeem	21039424	Database & Security Integration: Azure SQL Database Provisioning, EF Core Production Migrations, Seed Data & ASP.NET Identity Cookie Hardening
3	Habiba Khaled Fathallah Ebrahim	21132559	Course Module Integration: Course CRUD Service Wiring, Section/Lesson Hierarchy Validation & Static Asset Integration
4	Mariem Badwey Abdelsatar Ali	21171234	Media Player & Assessment Integration: Video.js Signed URL Playback Integration, AJAX Progress Sync & Quiz Engine Wiring
5	Ahmed Hany Mohamed Allam	21060002	Third-Party Cloud & Payment Integration: Azure.Storage.Blobs SDK, SAS Token Generator, Stripe.net Checkout/Webhooks & iTextSharp PDF Engine
6	Ahmed Mohammed Abdelfattah Mohammed	21147537	Frontend & Analytics Integration: Bootstrap 5 Responsive Bundle, Chart.js Dashboard Data Binding & Admin Approval Workflow Wiring

5.1 Comprehensive Technology Stack Specification

5.1.1 Architectural Stack Overview

The Lumora Learning Platform is engineered upon an enterprise-grade Microsoft .NET 8 ecosystem, combining server-side ASP.NET Core MVC (Razor Views) with cloud-native Azure PaaS services and industry-standard third-party SDKs. Every technology in the stack was selected to maximize full-stack development velocity, enforce strict role-based security, protect intellectual video property, and guarantee seamless deployment on Microsoft Azure.

Stack Layer	Technology / Library	Version / Tier	Role & Architectural Justification in Lumora
Backend Framework	ASP.NET Core MVC (C# 12)	.NET 8.0 LTS	Provides high-performance server-side Razor rendering, built-in Dependency Injection (IoC), routing, model binding, and unified full-stack deployment without CORS overhead.
Frontend UI & Layout	HTML5, CSS3, Bootstrap 5	Bootstrap v5.3	Delivers a mobile-first, responsive 12-column grid layout, accessible form controls, modals, and consistent design tokens across Student, Instructor, and Admin views.
Client Interactivity	JavaScript (ES6+) & AJAX	ECMAScript 2023	Handles asynchronous lesson completion updates, dynamic shopping cart interactions, and form validations without full-page reloads.
Video Streaming UI	Video.js Player	v8.x HTML5	Renders adaptive HTML5 video playback with custom controls, playback speed adjustment, and completion event listeners wired to Azure Signed URLs.
Data Visualization	Chart.js	v4.x Canvas	Renders interactive bar, line, and doughnut charts inside Instructor and Admin dashboards to visualize monthly revenue, top courses, and enrollment KPIs.
ORM & Persistence	Entity Framework Core	EF Core 8.0	Manages object-relational mapping, LINQ-to-SQL query translation, Fluent API schema configurations, automated migrations, and SQL connection resiliency.
Relational Database	Microsoft SQL Server / Azure SQL	SQL Server 2022 / Azure PaaS	Stores normalized (3NF) transactional data across Users, Courses, Sections, Lessons, Enrollments, Quizzes, Reviews, Payments, and Certificates with ACID compliance.
Authentication & RBAC	ASP.NET Core Identity	.NET 8.0 Identity	Manages PBKDF2 password hashing, encrypted HTTP-only authentication cookies, CSRF anti-forgery tokens, and Role-Based Access Control (Admin, Instructor, Student).

Cloud Video Storage	Azure Blob Storage & Azure CDN	Azure Storage v12 SDK	Stores raw MP4 course lectures in private cloud containers and generates time-limited Shared Access Signature (SAS) Signed URLs to prevent unauthorized scraping.
Payment Gateway	Stripe API (Stripe.net SDK)	Stripe API v2024+ (Test Mode)	Executes PCI-DSS compliant hosted checkout sessions, discount coupon deductions, and cryptographic HMAC webhook verification for instant enrollment activation.
Certificate Engine	iTextSharp / iText7 PDF	v5.5 / v8 PDF	Dynamically synthesizes tamper-evident PDF completion certificates stamped with student name, course title, issue date, and a unique database-backed Verification ID.
Version Control & CI/CD	Git, GitHub & GitHub Actions	Git 2.x / Actions v4	Hosts the monorepo (lumora-learning-platform), enforces pull-request reviews, runs automated xUnit builds/tests, and deploys artifacts to Azure App Services.

5.1.2 NuGet Packages & Client Library Manifest

To maintain strict N-Tier separation of concerns, external NuGet packages are installed only within the specific Visual Studio project assembly that requires them:

Target Project Assembly	NuGet Package / Library Identifier	Purpose & Integration Scope
Lumora.Core	Microsoft.Extensions.Identity.Stores	Provides base IdentityUser<string> contracts for ApplicationUser without coupling Core to EF Core SQL drivers.
Lumora.Data	Microsoft.EntityFrameworkCore.SqlServer Microsoft.EntityFrameworkCore.Tools Microsoft.AspNetCore.Identity.EntityFrameworkCore	Supplies SQL Server database provider, migration CLI tools, and IdentityDbContext<ApplicationUser> persistence tables.
Lumora.Services	Azure.Storage.Blobs Stripe.net iTextSharp.LGPLv2.Core	Encapsulates cloud blob upload/SAS generation, Stripe checkout session creation, and server-side PDF certificate rendering.
Lumora.Web	Microsoft.AspNetCore.Mvc.Razor.RuntimeCompilation Bootstrap 5.3 / Chart.js / Video.js (wwwroot/lib)	Enables Razor view compilation, Identity cookie middleware, Dependency Injection registration, and static client libraries.
Lumora.UnitTests	xunit / Moq / FluentAssertions	Supports isolated unit testing of Business Services, Pricing Strategies, and KPI calculations using mocked repositories.

5.2 Component Diagram & Software Dependencies

5.2.1 Component Architecture & Interface Contracts

In accordance with Clean N-Tier Architecture and the Dependency Inversion Principle (SOLID), Lumora's codebase is partitioned into four compiled .NET assemblies. High-level presentation components never depend directly on low-level data access implementations; instead, all cross-layer communication occurs through strongly-typed C# interfaces defined in Lumora.Core and resolved at runtime via ASP.NET Core's built-in Inversion of Control (IoC) container.

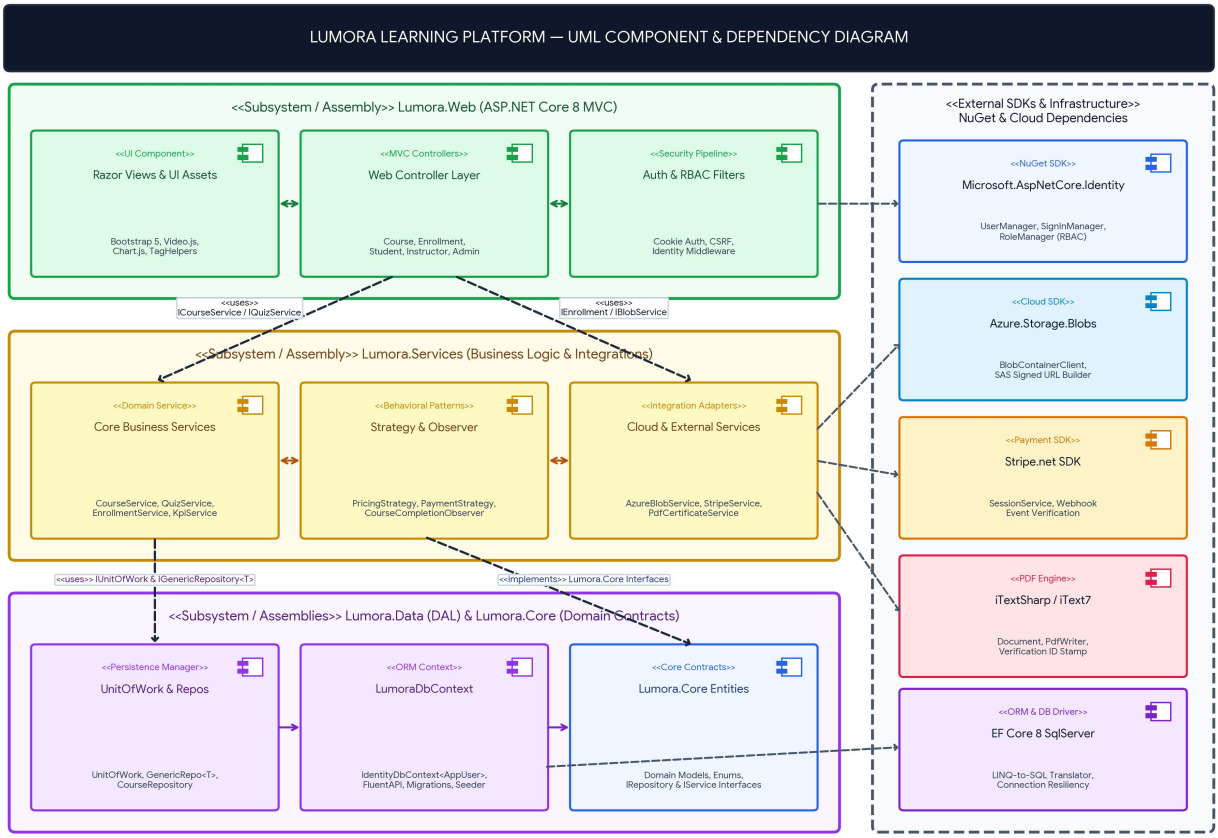


Figure 5.1: Lumora Learning Platform — UML Component & External Dependency Diagram

5.2.2 Provided & Required Component Interfaces Matrix

The table below details each software component, the interfaces it exposes (Provided Interfaces), the contracts it consumes (Required Interfaces), and its injected lifetime scope:

Component Name	Host Assembly	Provided Interface(s)	Required Interface(s) / SDKs	DI Lifetime
CourseService	Lumora.Services	ICourseService	IUnitOfWork, ICourseRepository, IAzureBlobService	Scoped
EnrollmentService	Lumora.Services	IEnrollmentService	IUnitOfWork, IPaymentStrategy,	Scoped

			IPricingStrategy	
QuizAssessmentService	Lumora.Services	IQuizService	IUnitOfWork, CourseCompletionSubject (Observer)	Scoped
KpiAnalyticsService	Lumora.Services	IKpiAnalyticsService	IUnitOfWork, ICourseRepository, IEnrollmentRepository	Scoped
AzureBlobStorageService	Lumora.Services	IAzureBlobService	Azure.Storage.Blobs (BlobServiceClient, BlobSasBuilder)	Singleton / Scoped
StripePaymentService	Lumora.Services	IPaymentStrategy	Stripe.net (SessionService, EventUtility)	Scoped
PdfCertificateService	Lumora.Services	IPdfCertificateService	iTextSharp Document Engine, IUnitOfWork	Scoped
UnitOfWork	Lumora.Data	IUnitOfWork	LumoraDbContext, IGenericRepository<T>	Scoped
GenericRepository<T>	Lumora.Data	IGenericRepository<T>	LumoraDbContext (DbSet<T>)	Scoped

5.3 Deployment Diagram & Cloud Infrastructure Topology

5.3.1 Physical & Cloud Runtime Architecture

Lumora is designed for cloud-native Platform-as-a-Service (PaaS) hosting on Microsoft Azure. Rather than managing raw virtual machines, the deployment topology leverages managed Azure services to guarantee high availability, automated SSL/TLS termination, elastic vertical/horizontal scaling, and isolated private storage for instructional videos.

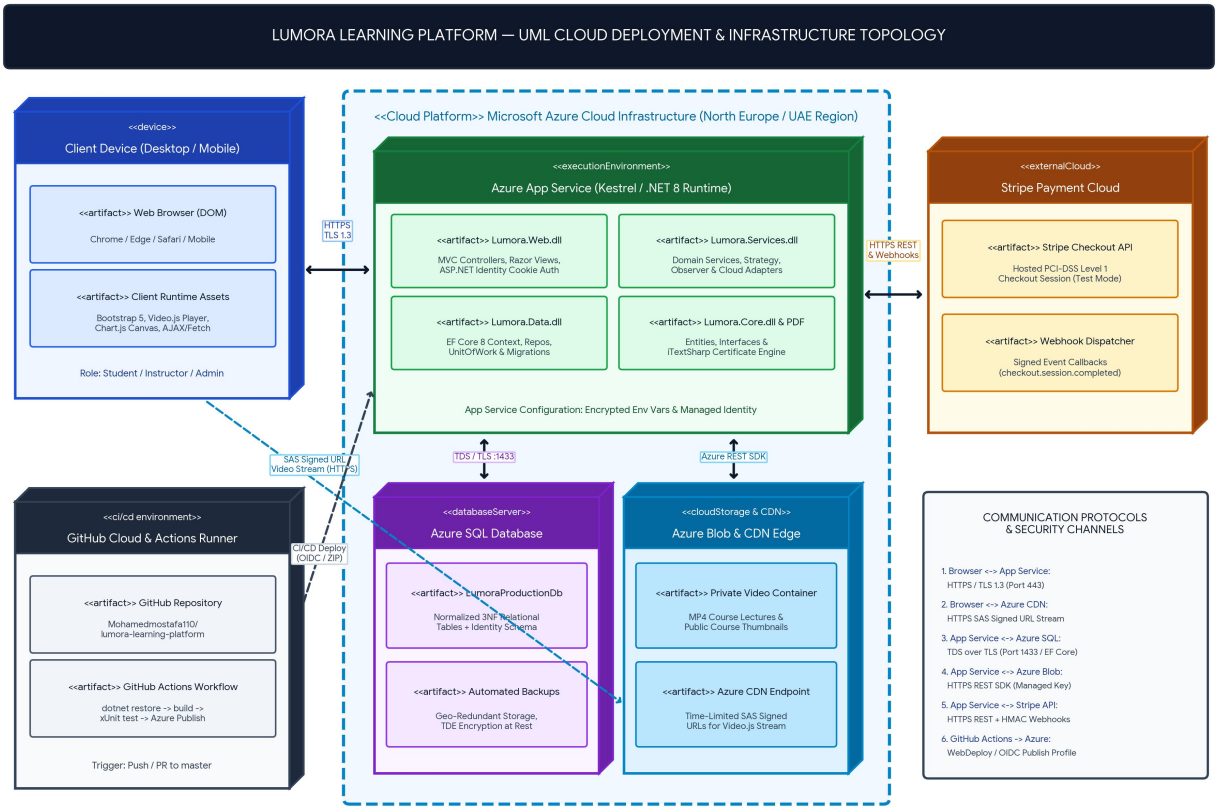


Figure 5.2: Lumora Learning Platform — UML Cloud Deployment & Hardware Distribution Diagram

5.3.2 Deployment Nodes & Network Security Specifications

Each deployment node in Figure 5.2 is governed by strict network protocols, firewall rules, and runtime configurations:

UML Node / Environment	Hosted Artifacts & Services	Network Protocol & Port	Security & Scaling Configuration
1. Client Device Node (Desktop / Mobile Browser)	HTML5 DOM, Bootstrap 5 CSS/JS, Video.js Player, Chart.js Canvas.	HTTPS / TLS 1.3 (TCP Port 443)	Enforces HSTS, Secure/HttpOnly Identity Cookies, SameSite=Lax, and Anti-Forgery CSRF validation tokens.
2. Azure App Service (.NET 8 Managed Runtime)	Lumora.Web.dll, Lumora.Services.dll, Lumora.Data.dll, Lumora.Core.dll, iTextSharp.dll.	HTTPS (443) Inbound TDS (1433) Outbound	Runs behind Azure Reverse Proxy with automated SSL certificates, Always-On warmup, and encrypted App Settings.
3. Azure SQL Database (Managed Relational Engine)	LumoraProductionDb (3NF Schema + ASP.NET Identity Tables + Indexes).	Tabular Data Stream (TDS) over TLS 1.2+ (Port 1433)	Protected by Azure SQL Firewall (Allow Azure Services Only), Transparent Data Encryption (TDE) at rest, and automated backups.
4. Azure Blob Storage & CDN	Private 'course-videos' container (MP4s) & Public	HTTPS REST API & SAS Signed URLs	Public anonymous access disabled on video container; Video.js streams

(Media Storage Node)	'course-thumbnails' container.	(443)	via time-limited HMAC-SHA256 SAS tokens (60-min expiry).
5. Stripe Payment Cloud (External PCI Node)	Stripe Checkout Session Engine & Webhook Event Dispatcher (Test Mode).	HTTPS REST API & Signed Webhooks (443)	Zero raw card data touches Lumora servers; webhook callbacks verified via Stripe-Signature HMAC secret.
6. GitHub Actions Runner (CI/CD Build Node)	Ubuntu-latest build agent executing dotnet build, xUnit tests, and WebDeploy.	HTTPS OIDC / Publish Profile (Port 443)	Triggered on push/PR to master; blocks deployment automatically if any compilation error or unit test failure occurs.

5.4 External Systems Integration & CI/CD Pipeline Mechanics

5.4.1 Deep-Dive Integration Workflows (Azure Blob, Stripe & iTextSharp)

Lumora integrates three mission-critical external subsystems inside the Business Logic Layer (Lumora.Services) using clean adapter interfaces:

- **1. Secure Video Streaming via Azure Blob & Signed URLs:** When an Instructor uploads an MP4 lecture, CourseController streams the IFormFile to AzureBlobStorageService. The service uploads the stream to a private container ('course-videos') using a GUID blob identifier and saves only the blob name in SQL Server. When an enrolled Student opens the Course Player, AzureBlobStorageService generates a BlobSasBuilder token valid for 60 minutes with Read-only permission (BlobSasPermissions.Read) and binds the signed URI to the Video.js player.
- **2. PCI-Compliant Checkout via Stripe.net SDK:** When a Student initiates checkout, EnrollmentController delegates to PricingStrategy (applying any valid coupon code) and invokes StripePaymentService. A Stripe Checkout Session is created in Test Mode with SuccessUrl and CancelUrl callbacks containing the Order/Course metadata. Upon payment confirmation, Lumora verifies the session status and commits both the Payment and Enrollment entities inside a single atomic UnitOfWork transaction.
- **3. Dynamic Verifiable Certificates via iTextSharp:** When a Student completes 100% of course lessons and passes all section quizzes, the CourseCompletionSubject (Observer Pattern) triggers PdfCertificateService. Using iTextSharp, the service renders a high-resolution vector PDF certificate in memory, embeds a unique alphanumeric Verification Code (e.g., LUM-2026-8F4A9C), persists the record in the Certificates table, and streams the PDF download to the student.

5.4.2 IoC Container Registration & Production Resiliency (Program.cs)

Below is the architectural configuration in Lumora.Web/Program.cs demonstrating how EF Core SQL connection resiliency, ASP.NET Core Identity RBAC, and all N-Tier repositories, strategies, and cloud adapters are wired together:

```
// Lumora.Web/Program.cs — Enterprise N-Tier Dependency Injection & Cloud Configuration
var builder = WebApplication.CreateBuilder(args);

// 1. Configure Azure SQL Database with EF Core Connection Resiliency (Auto-Retry on Transient Faults)
```

```
builder.Services.AddDbContext<LumoraDbContext>(options =>
    options.UseSqlServer(builder.Configuration.GetConnectionString("LumoraDbConnection"),
        sqlOptions => {
            sqlOptions.MigrationsAssembly("Lumora.Data");
            sqlOptions.EnableRetryOnFailure(maxRetryCount: 5, maxRetryDelay: TimeSpan.FromSeconds(10),
errorNumbersToAdd: null);
        }));

// 2. Configure ASP.NET Core Identity & Hardened RBAC Cookie Security
builder.Services.AddIdentity<ApplicationUser, IdentityRole>(options => {
    options.Password.RequireDigit = true;
    options.Password.RequiredLength = 8;
    options.User.RequireUniqueEmail = true;
})
.AddEntityFrameworkStores<LumoraDbContext>()
.AddDefaultTokenProviders();

builder.Services.ConfigureApplicationCookie(options => {
    options.Cookie.HttpOnly = true;
    options.Cookie.SecurePolicy = CookieSecurePolicy.Always;
    options.ExpireTimeSpan = TimeSpan.FromHours(8);
    options.LoginPath = "/Account/Login";
    options.AccessDeniedPath = "/Account/AccessDenied";
});

// 3. Register Data Access Layer (Unit of Work & Generic Repositories)
builder.Services.AddScoped(typeof(IGenericRepository<>), typeof(GenericRepository<>));
builder.Services.AddScoped<ICourseRepository, CourseRepository>();
builder.Services.AddScoped<IUnitOfWork, UnitOfWork>();

// 4. Register Business Services, Design Patterns (Strategy/Observer) & Cloud Adapters
builder.Services.AddScoped<ICourseService, CourseService>();
builder.Services.AddScoped<IEnrollmentService, EnrollmentService>();
builder.Services.AddScoped<IQuizService, QuizAssessmentService>();
builder.Services.AddScoped<IKpiAnalyticsService, KpiAnalyticsService>();
builder.Services.AddScoped<IAzureBlobService, AzureBlobStorageService>();
builder.Services.AddScoped<IPaymentStrategy, StripePaymentService>();
builder.Services.AddScoped<IPdfCertificateService, PdfCertificateService>();

builder.Services.AddControllersWithViews();
```

5.4.3 Automated CI/CD Pipeline & Deployment Strategy

To enable all 6 members of the Code Raiders team to develop in parallel without integration regressions, Lumora enforces a structured GitHub Flow and an automated 6-stage Continuous Integration / Continuous Deployment (CI/CD) pipeline:

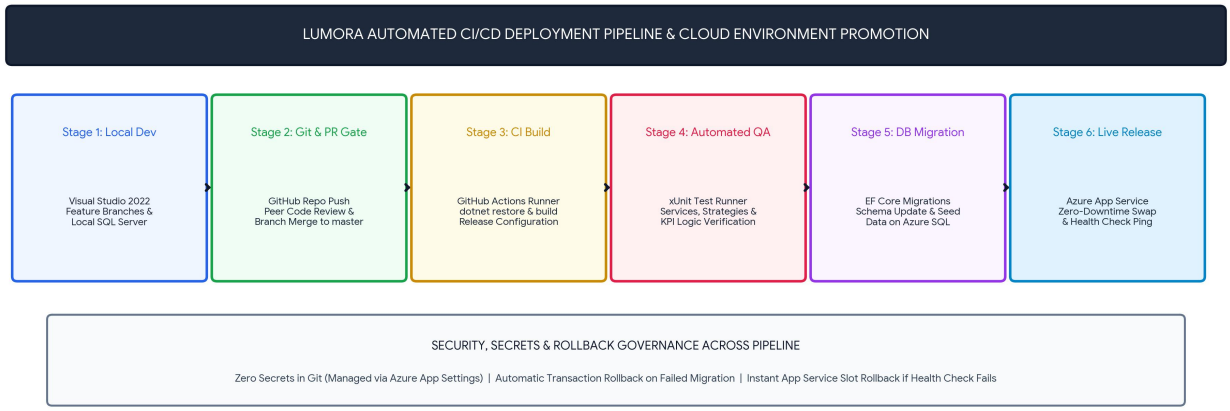


Figure 5.3: Lumora Learning Platform — 6-Stage Automated GitHub Actions CI/CD Pipeline

5.4.4 Environment Configuration & Secret Management Matrix

Sensitive API keys and connection strings are strictly excluded from the GitHub repository via .gitignore. During local development, secrets are stored in .NET User Secrets; in production, they are injected securely via Azure App Service Configuration environment variables:

Configuration Key	Local Development (Visual Studio)	Production Environment (Azure App Service)	Consuming Component
ConnectionStrings:LumoraDbConnection	LocalDB / SQL Server Express (Trusted_Connection=True)	Azure SQL Server Connection String (Encrypted TLS + Managed Identity)	LumoraDbContext (Lumora.Data)
AzureBlobSettings:ConnectionString	Azurite Local Storage Emulator / Dev Storage Account	Azure Storage Account Access Key (Injected via App Service Env Var)	AzureBlobStorageService
AzureBlobSettings:VideoContainer	course-videos-dev (Private Container)	course-videos-prod (Private Container + SAS Signing)	AzureBlobStorageService
StripeSettings:SecretKey	sk_test_51... (Developer Secret Store)	sk_test_51... (Encrypted App Service Application Setting)	StripePaymentService

StripeSettings:PublishableKey	pk_test_51... (appsettings.Development.json)	pk_test_51... (App Service Application Setting)	Enrollment Checkout View
StripeSettings:WebhookSecret	whsec_... (Stripe CLI Local Listener)	whsec_... (Production HTTPS Webhook Endpoint Secret)	StripeWebhookController